

Sex at maximum entropy: critical recombination potential in a multilocus spin model

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Abstract

Sex is commonly viewed as a process that increases genomic diversity, making it surprising that its potential advantage should be greatest at maximum entropy. We show how this occurs in a simple multilocus model. Selective degeneracy between competing genomes coincides with maximum configurational entropy, maximally negative aggregate covariance, and a divergent relaxation time. These effects produce a positive singularity in integrated recombination potential. The apparent paradox is resolved because configurational entropy and haplotype entropy describe different objects: maximum disorder among fitness effects can coexist with strong order in their packaging into whole genomes. An exact extension to arbitrarily many genotypes shows that, near a generic degeneracy of the leading fitness class, the dynamics reduce asymptotically to the same two-genotype critical theory.

Keywords: evolution of sex, recombination, entropy, linkage disequilibrium, criticality, statistical mechanics

1. Introduction

Sex and recombination are commonly associated with the generation of genetic diversity. Recombination breaks associations among loci and creates new haplotypes from existing alleles. It is therefore natural to think of sex as an entropy-increasing process. This intuition appears to conflict with a simple result developed here: in a binary representation of multilocus fitness differences, the potential advantage of recombination is greatest precisely at the maximum-entropy state. In the deterministic infinite-time limit, the integrated advantage diverges there.

The apparent contradiction arises because two distinct entropies are involved. The first measures the number of possible arrangements of locus-

specific fitness differences between competing genomes. We call this *configurational entropy*. The second is the Shannon entropy of the haplotypes actually present in the population [7]. We call this *haplotype entropy*. These entropies describe different probability spaces. Maximum disorder in the signs of fitness differences can therefore coexist with strong order in the way alleles are packaged into whole genomes.

This distinction matters because recombination acts on the latter organization. A population containing only two parental haplotypes can occupy the maximum-entropy macrostate of fitness-difference space while remaining far from maximum entropy in haplotype space. Recombination can then destroy information stored in linkage and generate many new haplotypes.

Negative associations among selected loci provide a classical route to an advantage of recombination [3, 2, 1, 6]. In the model below, the maximum-entropy state is also the state of maximally negative aggregate covariance. Moreover, it occurs when the total fitness difference between competing genomes vanishes, causing the lifetime of their association to diverge. The central structure is therefore

$$\begin{array}{c}
 \text{maximum configurational entropy} \\
 \Updownarrow \\
 S = 0 \\
 \left\{ \begin{array}{l} C_{\Sigma} = C_{\Sigma, \min} \quad (\text{maximally negative covariance}), \\ \tau \rightarrow \infty \quad (\text{critical slowing down}). \end{array} \right.
 \end{array} \tag{1}$$

Together, the two consequences of selective degeneracy produce a positive singularity in recombination potential.

Our use of a spin representation is mathematical rather than physical. Binary spins provide a convenient coordinate system in which the combinatorial origin of the singularity becomes transparent [4]. The resulting theory connects distance from maximum configurational entropy directly to the cumulative evolutionary potential of recombination.

The initial derivation considers two competing genomes. We then remove this restriction. For an arbitrary number of genotypes, aggregate covariance admits an exact decomposition into contributions from genotype pairs. Near a generic fitness degeneracy at the leading edge of the population, lower-fitness genotypes disappear on finite time scales and the slow dynamics reduce asymptotically to those of the two leading genotypes. The two-genotype model is therefore not only a minimal example; it is the effective critical

theory of a generic pairwise fitness degeneracy in a many-genotype population.

2. Two genomes as a spin system

Consider two haploid genomes that differ at L additive fitness loci. Let

$$\Delta X_i = X_{2i} - X_{1i} \quad (2)$$

denote the contribution of locus i to the fitness difference between genome 2 and genome 1. We first consider the binary limit

$$\Delta X_i = \sigma_i, \quad \sigma_i \in \{-1, +1\}. \quad (3)$$

The total fitness difference between the two genomes is

$$S = \sum_{i=1}^L \sigma_i. \quad (4)$$

Thus S is simultaneously the selective difference between the genomes and the magnetization of the corresponding spin configuration.

If n_+ locus effects favor genome 2 and n_- favor genome 1, then

$$n_+ = \frac{L + S}{2}, \quad n_- = \frac{L - S}{2}. \quad (5)$$

For even L , selective degeneracy occurs at $S = 0$, where exactly half of the component effects favor each genome. Importantly, $S = 0$ does not require the component effects themselves to be small. The genomes can differ strongly at every locus while having identical total fitness because their component differences cancel.

3. Maximum entropy and the apparent paradox

3.1. Configurational entropy

For fixed L and S , the number of spin configurations compatible with the same total fitness difference is

$$\Omega_L(S) = \binom{L}{(L + S)/2}. \quad (6)$$

We define the corresponding configurational entropy as

$$\mathcal{H}(S) = \log \Omega_L(S). \quad (7)$$

This entropy measures how many different arrangements of locus-specific advantages and disadvantages produce the same total fitness difference.

The multiplicity is greatest at $S = 0$, so

$$S = 0 \iff \mathcal{H} = \mathcal{H}_{\max}. \quad (8)$$

For large L , expansion around $S = 0$ gives

$$\mathcal{H}_{\max} - \mathcal{H}(S) = \frac{S^2}{2L} + O\left(\frac{S^4}{L^3}\right). \quad (9)$$

Near the entropy maximum, the squared total fitness difference is therefore a direct measure of distance from the maximum-entropy state.

3.2. The maximum-entropy paradox

Equation (8) creates an apparent paradox. If sex acts by increasing genomic diversity, why should its potential advantage be greatest where entropy is already maximal?

The resolution is that $\mathcal{H}$ is not the entropy of the population's haplotypes. It is an entropy over possible patterns of fitness differences between the two parental genomes. To see the distinction, suppose the population initially consists of exactly two parental haplotypes, each at frequency one half, and that the parents differ at every one of the L loci. The Shannon entropy of complete haplotypes is then

$$H_{\text{hap}} = - \sum_g p_g \log p_g = \log 2. \quad (10)$$

There are only two genomes in the population.

Now choose the signs of their locus-specific fitness differences so that half are $+1$ and half are -1 . Then $S = 0$, and consequently $\mathcal{H} = \mathcal{H}_{\max}$. The same population therefore satisfies

$$\mathcal{H} = \mathcal{H}_{\max} \quad \text{while} \quad H_{\text{hap}} = \log 2. \quad (11)$$

Maximum entropy in fitness-difference space can therefore coexist with very low entropy in haplotype space.

This distinction is central. The statement that sex is most advantageous at maximum entropy does not mean that recombination somehow pushes an already maximal entropy still higher. Rather, maximum entropy in the space of fitness-difference configurations identifies a state in which extensive organization can remain in a different space: the association of alleles into whole haplotypes.

3.3. Latent haplotype entropy

The distinction becomes especially clear by examining alleles rather than whole genomes. Because the two parental haplotypes occur at equal frequencies and differ at every locus,

$$H(A_i) = \log 2 \quad (12)$$

for every locus. Hence

$$\sum_{i=1}^L H(A_i) = L \log 2. \quad (13)$$

Nevertheless, the entropy of the complete haplotype remains only

$$H(A_1, \dots, A_L) = \log 2, \quad (14)$$

because knowing the allele at any one locus identifies the complete parental genome.

The difference

$$\mathcal{I} = \sum_i H(A_i) - H(A_1, \dots, A_L) \quad (15)$$

is the total correlation, or multi-information [8]. For the two-parent population,

$$\mathcal{I}_{\text{parental}} = (L - 1) \log 2. \quad (16)$$

Thus the population contains an extensive amount of information stored in associations among loci.

Under complete independent assortment, while retaining allele frequencies of one half, the possible haplotypes number 2^L . At linkage equilibrium,

$$H_{\text{hap}}^{\text{LE}} = L \log 2, \quad \mathcal{I}^{\text{LE}} = 0. \quad (17)$$

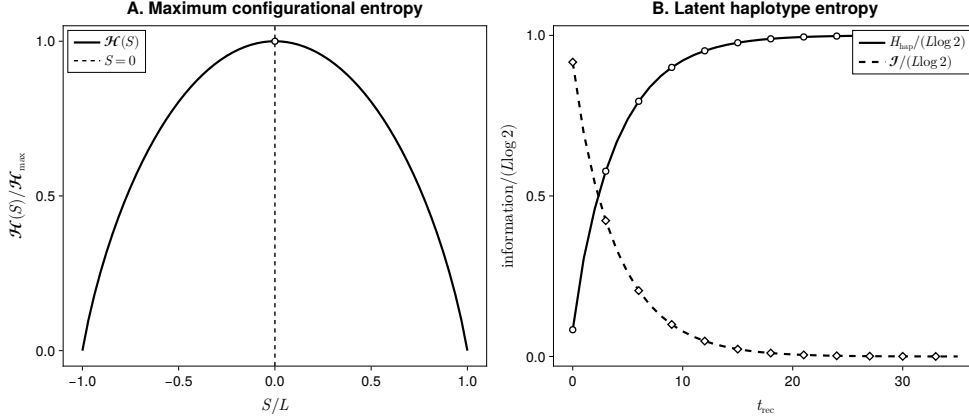


Figure 1: Maximum configurational entropy does not imply maximum haplotype entropy. (A) Configurational entropy of binary fitness-difference configurations peaks at $S = 0$. (B) Starting from two complementary parental haplotypes, recombination can increase haplotype entropy while decreasing total correlation, thereby converting linkage information into diversity among whole haplotypes.

Ideal recombination can therefore produce the transformation

$$H_{\text{hap}} : \log 2 \longrightarrow L \log 2, \quad \mathcal{I} : (L - 1) \log 2 \longrightarrow 0. \quad (18)$$

Figure 1 summarizes this distinction. Configurational entropy peaks at $S = 0$, whereas a population restricted to two parental haplotypes remains far below the maximum haplotype entropy permitted by its allele frequencies. Recombination can therefore increase haplotype entropy while configurational entropy is already maximal.

The population can thus be maximally disordered in one sense and highly ordered in another. At $S = 0$, the signs of the locus-specific fitness differences have maximum configurational entropy, whereas the alleles carrying those effects can remain perfectly associated into only two parental backgrounds. The two-parent population consequently contains considerable *latent haplotypic diversity*: many combinations are permitted by its allele frequencies but remain unrealized because the alleles are locked into parental genomes.

4. Maximum entropy maximizes negative covariance

The entropy maximum is not merely combinatorially common. It also has a special relationship to the genomic associations on which recombination acts.

Let genome 2 occur at frequency p and genome 1 at frequency $1 - p$. For loci i and j ,

$$\text{cov}(X_i, X_j) = p(1 - p)\sigma_i\sigma_j. \quad (19)$$

Define the aggregate pairwise covariance

$$C_\Sigma = \sum_{i < j} \text{cov}(X_i, X_j). \quad (20)$$

Using

$$\left(\sum_i \sigma_i \right)^2 = \sum_i \sigma_i^2 + 2 \sum_{i < j} \sigma_i \sigma_j \quad (21)$$

together with $\sigma_i^2 = 1$ gives

$$\sum_{i < j} \sigma_i \sigma_j = \frac{S^2 - L}{2}. \quad (22)$$

Consequently,

$$C_\Sigma = \frac{p(1 - p)}{2}(S^2 - L). \quad (23)$$

For fixed p , C_Σ reaches its minimum at $S = 0$:

$$C_\Sigma(0) = -\frac{L}{2}p(1 - p). \quad (24)$$

Maximum configurational entropy therefore coincides exactly with maximally negative aggregate covariance.

The reason is straightforward. Equal total fitness requires positive and negative locus effects to compensate. The configuration with greatest combinatorial multiplicity is therefore also the configuration with the strongest aggregate antagonism among component fitness effects.

The effect is genuinely collective. At $S = 0$,

$$\frac{\sum_{i < j} \sigma_i \sigma_j}{\binom{L}{2}} = -\frac{1}{L - 1}. \quad (25)$$

The mean association of any particular pair of loci therefore approaches zero as L grows. There are, however, $O(L^2)$ locus pairs, so these individually weak associations sum to an $O(L)$ aggregate effect.

5. Recombination potential and critical slowing down

We define the instantaneous aggregate recombination-potential rate by

$$R(t) = -2C_{\Sigma}(t), \quad (26)$$

which is the equal-weight pairwise extension of the two-locus covariance criterion. Equation (23) gives

$$R(t) = p(t)[1 - p(t)](L - S^2). \quad (27)$$

For equal initial parental frequencies, $p(0) = 1/2$,

$$R(0) = \frac{L - S^2}{4}, \quad (28)$$

and is maximal at $S = 0$.

Maximum configurational entropy therefore already maximizes the instantaneous quantity favoring breakup of the parental associations. The effect becomes singular when time is included.

Under deterministic selection,

$$\frac{dp}{dt} = Sp(1 - p). \quad (29)$$

For $p(0) = 1/2$,

$$p(t) = \frac{1}{2} \left[1 + \tanh \left(\frac{St}{2} \right) \right]. \quad (30)$$

The relaxation rate is $|S|$, so the characteristic relaxation time scales as

$$\tau \sim |S|^{-1}. \quad (31)$$

At $S = 0$, the two genomes have identical total fitness and $dp/dt = 0$ for every p . Their deterministic association therefore persists indefinitely.

The same condition that makes aggregate covariance maximally negative consequently makes that covariance arbitrarily long lived. This coincidence is the origin of the singular cumulative effect.

6. A positive singularity at maximum entropy

Define the integrated recombination potential of a complete selective episode as

$$V_L = -2 \int_0^\infty C_\Sigma(t) dt. \quad (32)$$

For $S \neq 0$ and $p(0) = 1/2$,

$$\int_0^\infty p(t)[1 - p(t)] dt = \frac{1}{2|S|}. \quad (33)$$

It follows that

$$V_L = \frac{L - S^2}{2|S|} = \frac{L}{2|S|} - \frac{|S|}{2}. \quad (34)$$

Near selective degeneracy,

$$V_L \sim \frac{L}{2|S|}. \quad (35)$$

The divergence is necessarily positive. Approaching $S = 0$ while retaining nonzero component differences requires cancellation among opposing locus effects, precisely the configuration that gives negative aggregate covariance. The critical state is therefore neutral with respect to *total parental fitness*, but not with respect to recombination. Its internal genomic structure is strongly oriented in the direction that favors breakup of associations.

Sex potential becomes large for two reasons generated by the same condition: the genomic association has the favorable sign and it persists for an anomalously long time. In compact form,

$$S \rightarrow 0 \implies C_\Sigma \rightarrow C_{\Sigma, \min} \quad \text{and} \quad \tau \rightarrow \infty, \quad (36)$$

which together imply $V_L \rightarrow +\infty$ in the deterministic infinite-time limit.

7. Critical scaling with entropy

Equation (9) gives

$$\mathcal{H}_{\max} - \mathcal{H} \simeq \frac{S^2}{2L}, \quad (37)$$

and therefore

$$|S| \simeq \sqrt{2L(\mathcal{H}_{\max} - \mathcal{H})}. \quad (38)$$

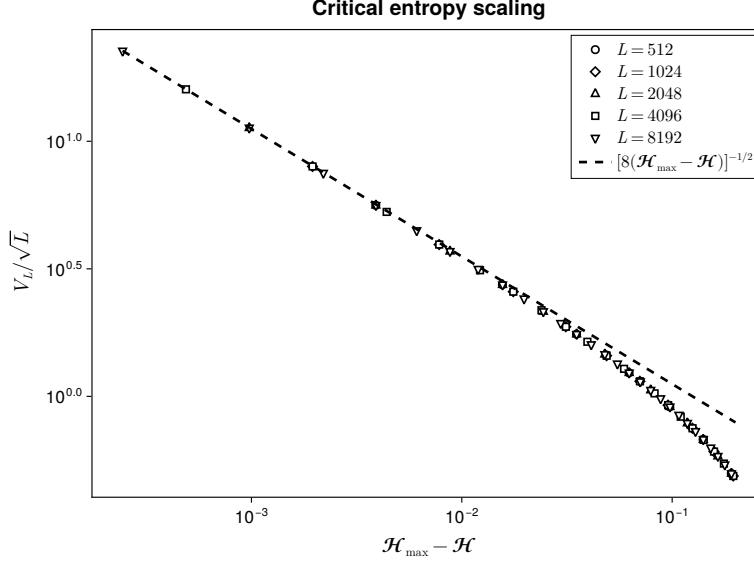


Figure 2: Critical scaling of recombination potential with entropy deficit. Near the maximum-entropy state, the theory predicts $V_L/\sqrt{L} \sim [8(\mathcal{H}_{\max} - \mathcal{H})]^{-1/2}$. Points show exact finite- L values; the dashed line shows the asymptotic prediction.

Substitution into equation (35) yields the central scaling relation

$$V_L \sim \sqrt{\frac{L}{8}} (\mathcal{H}_{\max} - \mathcal{H})^{-1/2}. \quad (39)$$

Equivalently,

$$\frac{V_L}{\sqrt{L}} \sim \frac{1}{\sqrt{8(\mathcal{H}_{\max} - \mathcal{H})}}. \quad (40)$$

Integrated recombination potential therefore diverges with exponent $1/2$ as the fitness-difference configuration approaches maximum configurational entropy. The divergence occurs not as this entropy is being created, but as it approaches its maximum. Figure 2 shows the exact finite- L relation and its convergence toward the predicted inverse-square-root form.

The result becomes intuitive once the levels of description are separated. Configurational entropy measures disorder among possible patterns of fitness effects. Recombination instead destroys information stored in associations among loci and thereby increases the diversity of realized haplotypes. Max-

imum entropy in one space can therefore define the condition under which entropy production in another space has its greatest evolutionary potential.

8. Continuous criticality and finite-time regularization

Define the scaled distance from the symmetric state by

$$\phi = \frac{|S|}{\sqrt{L}}. \quad (41)$$

Near the entropy maximum,

$$\mathcal{H}_{\max} - \mathcal{H} \simeq \frac{\phi^2}{2}, \quad (42)$$

so

$$\phi \sim (\mathcal{H}_{\max} - \mathcal{H})^{1/2}. \quad (43)$$

The distance from the symmetric state therefore vanishes continuously with square-root exponent $1/2$, whereas the integrated response $V_L/\sqrt{L}$ diverges with the reciprocal exponent.

The point $S = 0$ is also dynamically critical because its relaxation rate vanishes and its relaxation time diverges. We use the terms *continuous critical point* and *second-order criticality* in this dynamical scaling sense. A conventional equilibrium phase transition would additionally require an independently specified thermodynamic control parameter and a corresponding nonanalytic thermodynamic potential [5, 9]. The present analysis does not require that stronger identification.

Direct numerical integration provides a simple check on the exact expression in equation (34). Figure 3A compares numerical and exact values over a range of positive and negative S . Figure 3B illustrates an important regularization of the infinite-time singularity. At exact degeneracy, $p(t) = 1/2$ and

$$V_L(T) = \frac{LT}{4}. \quad (44)$$

Thus finite observation time replaces the infinite-time divergence by linear growth with T .

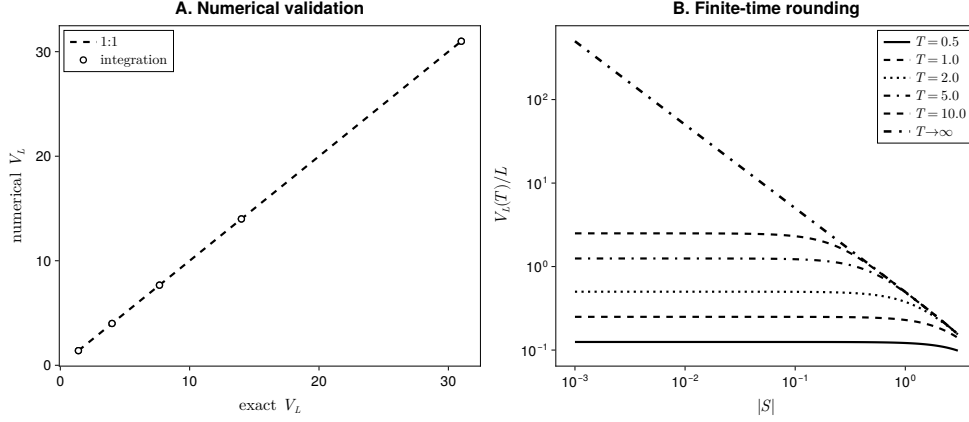


Figure 3: Exact dynamical validation and finite-time rounding. (A) Direct numerical integration of the deterministic dynamics agrees with the exact expression for V_L . (B) Finite observation time rounds the $S = 0$ singularity; longer times approach the infinite-time critical form.

9. Continuous fitness effects and the ensemble tail

The binary model makes the entropy calculation transparent but leaves S discretely spaced. A natural extension permits locus-specific magnitudes to vary:

$$\Delta X_i = a_i \sigma_i, \quad a_i > 0. \quad (45)$$

Define

$$S = \sum_i a_i \sigma_i, \quad A_2 = \sum_i a_i^2. \quad (46)$$

The same quadratic identity gives

$$\sum_{i < j} a_i a_j \sigma_i \sigma_j = \frac{S^2 - A_2}{2}, \quad (47)$$

and hence

$$V = \frac{A_2 - S^2}{2|S|}. \quad (48)$$

Near selective degeneracy,

$$V \sim \frac{A_2}{2|S|}. \quad (49)$$

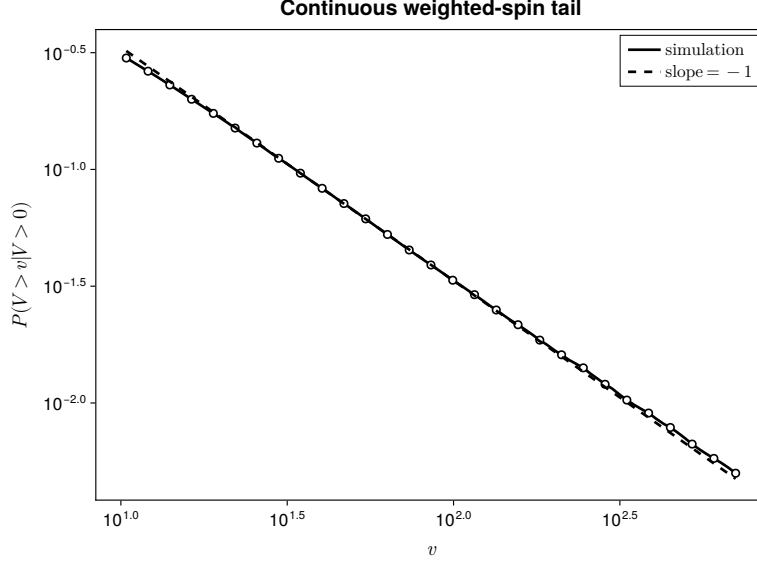


Figure 4: Continuous weighted-spin extension. Allowing continuously varying locus-effect magnitudes removes the discrete spacing of S and yields the predicted $\Pr(V > v \mid V > 0) \sim v^{-1}$ tail.

If the density of S is smooth and nonzero near zero, large positive values of V correspond to a region of width proportional to $1/v$ around the critical manifold. Consequently,

$$\Pr(V > v) \sim v^{-1}. \quad (50)$$

Figure 4 illustrates this ensemble-level consequence using continuously distributed locus-effect magnitudes.

The same argument applies in the many-genotype setting when the relevant variable is the gap δ between the two leading genotype fitnesses. If the density of this leading gap is finite and nonzero near zero, equation (74) again yields an inverse-power tail. Thus the local critical singularity and the ensemble-level heavy tail have the same origin: near-degeneracy of leading genomic backgrounds.

10. Extension to multiple competing genotypes

The preceding results were derived for a population containing only two genotypes. We now show that the central structure survives when the population contains an arbitrary number of competing genotypes.

10.1. Exact pairwise decomposition

Let $g = 1, \dots, K$ index genotypes and $i = 1, \dots, L$ index additive fitness components. Genotype g has component values X_{gi} and total fitness

$$W_g = \sum_{i=1}^L X_{gi}. \quad (51)$$

Let p_g denote its population frequency, with $\sum_g p_g = 1$. Under deterministic selection,

$$\frac{dp_g}{dt} = p_g(W_g - \bar{W}), \quad \bar{W} = \sum_g p_g W_g. \quad (52)$$

For a pair of genotypes g and h , define the component differences

$$\Delta X_i^{gh} = X_{gi} - X_{hi}, \quad (53)$$

their total fitness difference

$$S_{gh} = W_g - W_h = \sum_i \Delta X_i^{gh}, \quad (54)$$

and their squared distance in component-fitness space

$$Q_{gh} = \sum_i \left(\Delta X_i^{gh} \right)^2. \quad (55)$$

For any two component traits X_i and X_j , covariance across the population has the pairwise representation

$$\text{cov}(X_i, X_j) = \frac{1}{2} \sum_{g,h} p_g p_h \Delta X_i^{gh} \Delta X_j^{gh}. \quad (56)$$

Equivalently, summing over unordered genotype pairs,

$$\text{cov}(X_i, X_j) = \sum_{g < h} p_g p_h \Delta X_i^{gh} \Delta X_j^{gh}. \quad (57)$$

Summing over locus pairs gives

$$C_\Sigma = \sum_{g < h} p_g p_h \sum_{i < j} \Delta X_i^{gh} \Delta X_j^{gh}. \quad (58)$$

For each genotype pair,

$$S_{gh}^2 = Q_{gh} + 2 \sum_{i < j} \Delta X_i^{gh} \Delta X_j^{gh}, \quad (59)$$

and therefore

$$C_\Sigma = \frac{1}{2} \sum_{g < h} p_g p_h (S_{gh}^2 - Q_{gh}). \quad (60)$$

The corresponding aggregate recombination-potential rate is

$$R = -2C_\Sigma = \sum_{g < h} p_g p_h (Q_{gh} - S_{gh}^2). \quad (61)$$

Equation (61) is exact for arbitrary K . The many-genotype recombination potential is therefore a frequency-weighted sum of contributions from genotype pairs. The two-genotype theory is the elementary term in this exact decomposition.

Defining

$$I_{gh} = \int_0^\infty p_g(t) p_h(t) dt, \quad (62)$$

the integrated many-genotype potential becomes

$$V_K = \sum_{g < h} (Q_{gh} - S_{gh}^2) I_{gh}. \quad (63)$$

Unlike the instantaneous decomposition, the quantities I_{gh} depend on all genotypes in the population. A pair can be nearly degenerate in fitness yet contribute little to the long-time potential if both members are rapidly eliminated by a substantially fitter genotype. The singular behavior is therefore controlled by near-degeneracy at the *leading edge* of the fitness distribution.

10.2. Reduction to the two leading genotypes

Order the genotype fitnesses as

$$W_1 \geq W_2 \geq W_3 \geq \cdots \geq W_K \quad (64)$$

and define the fitness gap between the two leading genotypes by

$$\delta = W_1 - W_2. \quad (65)$$

Suppose

$$\delta \rightarrow 0^+, \quad (66)$$

while the gap separating the second and third genotypes remains finite:

$$\Gamma = W_2 - W_3 > 0. \quad (67)$$

Under equation (52), relative genotype frequencies satisfy

$$\frac{p_g(t)}{p_h(t)} = \frac{p_g(0)}{p_h(0)} e^{(W_g - W_h)t}. \quad (68)$$

Genotypes $3, \dots, K$ therefore disappear relative to the leading pair on time scales of order Γ^{-1} or shorter. These time scales remain finite as $\delta \rightarrow 0$. By contrast, the relative frequency of the two leading genotypes changes on the diverging time scale

$$\tau_{\text{slow}} \sim \delta^{-1}. \quad (69)$$

After the finite transient in which lower-fitness backgrounds disappear, the remaining slow dynamics are therefore those of the leading pair. Selection itself performs the reduction from a K -genotype system to an effective two-genotype system.

This reduction can be made quantitative. Let the initial frequencies of the two leading genotypes be p_1^0 and p_2^0 . In the limit $\delta \rightarrow 0^+$ with Γ fixed,

$$I_{12} = \int_0^\infty p_1(t)p_2(t) dt \sim \frac{c_{12}}{\delta}, \quad (70)$$

where

$$c_{12} = \frac{p_2^0}{p_1^0 + p_2^0}. \quad (71)$$

The finite transient generated by the remaining genotypes contributes only $O(1)$ corrections to equation (70). For equal initial genotype frequencies, $p_g(0) = 1/K$, we have $c_{12} = 1/2$ for every K , so

$$I_{12} \sim \frac{1}{2\delta}. \quad (72)$$

The singular part of equation (63) is consequently

$$V_K^{\text{sing}} \sim \frac{c_{12}(Q_{12} - \delta^2)}{\delta}, \quad (73)$$

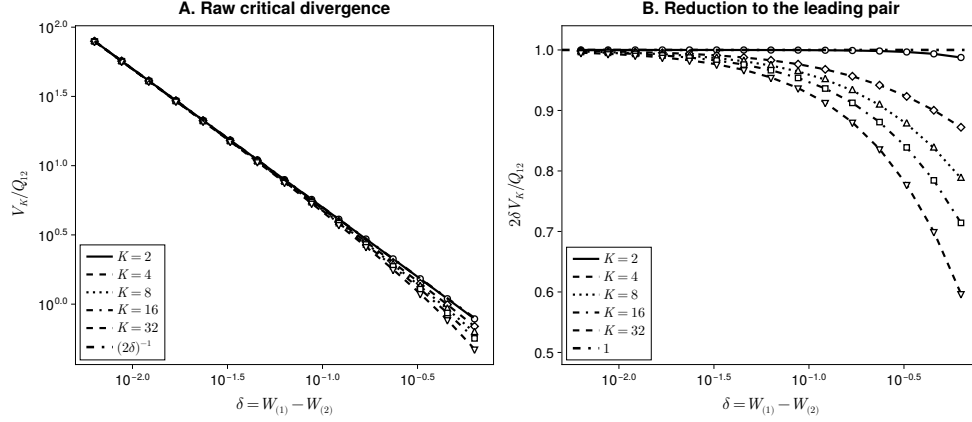


Figure 5: Many-genotype critical reduction. (A) The raw K -genotype divergence increases as the leading fitness gap δ approaches zero. (B) After normalization, simulations for different K collapse toward $2\delta V_K/Q_{12} = 1$, demonstrating reduction to the two leading genotypes.

or, near degeneracy,

$$V_K^{\text{sing}} \sim \frac{c_{12}Q_{12}}{\delta}. \quad (74)$$

All contributions involving genotypes separated from the fitness maximum by a finite gap remain finite and form a nonsingular background. For equal initial frequencies,

$$V_K^{\text{sing}} \sim \frac{Q_{12}}{2\delta}. \quad (75)$$

The two-genotype singularity therefore survives unchanged in form when arbitrarily many genotypes are present.

Figure 5 tests this prediction directly. The unscaled potential diverges as the leading fitness gap closes, whereas the normalized quantity $2\delta V_K/Q_{12}$ approaches unity across different values of K .

10.3. Entropy scaling of the leading pair

The configurational-entropy interpretation also survives. Suppose the two leading genotypes differ at d_{12} loci, and that their nonzero component differences have equal magnitude a :

$$\Delta X_i^{12} \in \{-a, +a\} \quad (76)$$

at the loci at which they differ. Then

$$Q_{12} = a^2 d_{12}. \quad (77)$$

The number of sign configurations producing a specified total difference δ is

$$\Omega_{12}(\delta) = \binom{d_{12}}{(d_{12} + \delta/a)/2}. \quad (78)$$

Near pairwise selective degeneracy,

$$\mathcal{H}_{12}^{\max} - \mathcal{H}_{12} \simeq \frac{\delta^2}{2Q_{12}}. \quad (79)$$

Hence

$$\delta \simeq \sqrt{2Q_{12} (\mathcal{H}_{12}^{\max} - \mathcal{H}_{12})}. \quad (80)$$

Combining this result with equation (74) gives

$$V_K^{\text{sing}} \sim c_{12} \sqrt{\frac{Q_{12}}{2}} (\mathcal{H}_{12}^{\max} - \mathcal{H}_{12})^{-1/2}. \quad (81)$$

For equal initial genotype frequencies, $c_{12} = 1/2$, and

$$V_K^{\text{sing}} \sim \sqrt{\frac{Q_{12}}{8}} (\mathcal{H}_{12}^{\max} - \mathcal{H}_{12})^{-1/2}. \quad (82)$$

The critical exponent is therefore unchanged by increasing the number of genotypes. Only the amplitude is modified, with Q_{12} replacing the fixed binary value L . Figure 6 expresses the many-genotype simulations in the corresponding entropy coordinate and recovers the same inverse-square-root scaling.

The appropriate entropy statement in a many-genotype population is consequently more precise than in the original two-genotype formulation: the singular advantage of recombination occurs near maximum configurational entropy of the fitness differences between the *leading competing genomic backgrounds*. It does not require maximum Shannon entropy of the population as a whole.

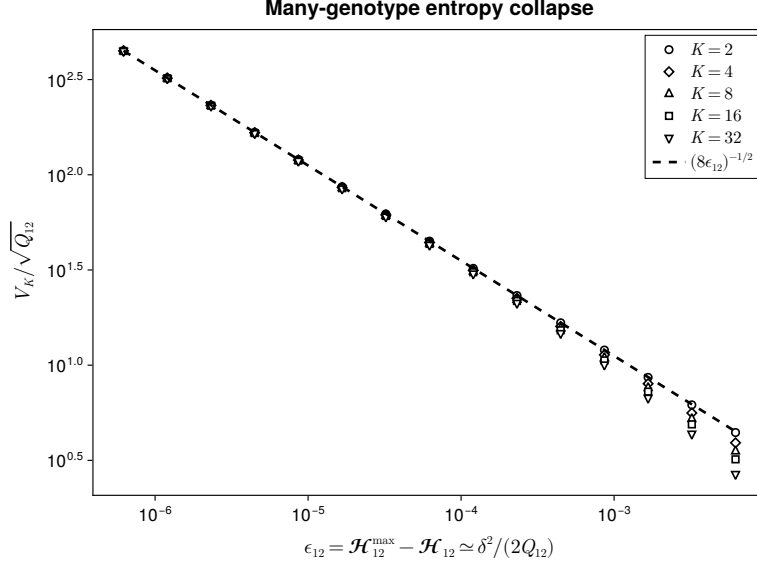


Figure 6: Many-genotype entropy-coordinate collapse. Expressed in the pairwise entropy coordinate $\epsilon_{12} \simeq \delta^2/(2Q_{12})$, the K -genotype simulations recover the same inverse-square-root singularity as the two-genotype theory.

10.4. Why pairwise criticality is generic

A two-way fitness degeneracy requires a single condition,

$$W_1 - W_2 = 0. \quad (83)$$

A three-way degeneracy requires two independent conditions,

$$W_1 - W_2 = 0, \quad W_1 - W_3 = 0, \quad (84)$$

and an m -way degeneracy generally requires $m - 1$ independent equalities. Pairwise degeneracies are therefore the generic critical events encountered under continuous variation of fitness parameters, whereas higher-order degeneracies occupy sets of higher codimension.

This observation explains why the two-genotype critical theory remains relevant in populations containing many genotypes. Near a generic level crossing at the fitness maximum, one slow direction appears while all other genotype-frequency directions retain finite relaxation rates. The system reduces to the dynamics of the two nearly degenerate leading backgrounds. Multiway degeneracies may produce different critical behavior and are not analyzed here.

11. Discussion

The central result is simple but initially counterintuitive: recombination potential is greatest at the maximum-entropy macrostate of locus-specific fitness differences. Sex is usually associated with the creation of genetic diversity, whereas a maximum-entropy state would seem to leave no further disorder to create.

The contradiction disappears because entropy is not an intrinsic scalar property of a genome or population. It is defined relative to a probability distribution over specified states. Here, configurational entropy counts possible arrangements of fitness differences between competing genomes, whereas haplotype entropy describes the distribution of whole genomes within the population. The first can be maximal while the second remains extremely small.

At $S = 0$, the signs of locus-specific fitness differences are maximally disordered in the combinatorial sense, but the alleles carrying those effects may still be confined to only two parental genomes. The population can therefore contain maximum disorder among fitness effects together with extensive order in their genomic packaging. In the two-parent example, this packaging contains $(L - 1) \log 2$ units of total correlation. Recombination destroys that information and can release the latent combinatorial diversity of the segregating alleles, raising haplotype entropy from $\log 2$ toward $L \log 2$.

The appropriate statement is therefore not simply that sex increases entropy. More precisely, sex destroys association information at one level of genomic organization and thereby permits greater diversity at another. Its strongest potential advantage can occur at maximum configurational entropy precisely because considerable order remains in the packaging of alleles into haplotypes.

The maximum-entropy state is also special dynamically. Equal numbers of opposing fitness effects make aggregate covariance maximally negative, while cancellation among those effects makes the total selective difference between the competing genomes vanish. Hence the genomic association favorable to recombination is strongest exactly where its lifetime becomes longest. Their combination yields

$$V_L \propto (\mathcal{H}_{\max} - \mathcal{H})^{-1/2}. \quad (85)$$

The sign and divergence therefore have a common mathematical origin: cancellation among opposing locus-specific fitness effects.

The extension to arbitrary genotype number strengthens this interpretation. Aggregate covariance admits the exact pairwise decomposition in equation (60). A nearly neutral pair contributes strongly only if it remains present long enough to matter, so the relevant degeneracies occur at the leading edge of the fitness distribution. When the fitness gap δ between the two leading genotypes approaches zero while the third-ranked genotype remains separated by a finite gap, all lower-fitness backgrounds decay on finite time scales. The only diverging time scale belongs to the leading pair.

The many-genotype system therefore undergoes a natural dynamical reduction,

$$\begin{array}{ccc}
 K \text{ genotypes} & & \\
 \downarrow & & \\
 \text{leading fitness class} & & (86) \\
 \downarrow & & \\
 \text{near-degenerate leading pair} & &
 \end{array}$$

Near a generic pairwise level crossing, the two-genotype calculation is consequently not merely a convenient approximation. It is the asymptotically dominant theory.

This result also clarifies the relevant entropy in a genetically diverse population. The singularity is associated with maximum configurational entropy of the component fitness differences between the leading competing genomic backgrounds, not with maximum population-wide haplotype entropy. Indeed, a population at complete linkage equilibrium would contain little association information for recombination to remove. The entropy paradox therefore survives increasing genotype number, but its interpretation becomes more precise.

Pairwise degeneracy is also expected to dominate generically. Equality of two leading fitnesses imposes one condition; equality of three imposes two; equality of m imposes $m - 1$. Higher-order degeneracies may therefore possess interesting critical properties but require additional tuning. Their analysis is beyond the present model and provides a natural extension of the theory.

Several limitations remain. The quantity $R = -2C_\Sigma$ assigns equal weight to all pairwise locus associations. Real recombination fractions vary with genomic distance and chromosome structure, so a more general model should weight covariance terms by the probability that recombination actually separates the relevant loci. Epistasis can also alter both the fitness geometry and the consequences of generating recombinant genotypes. Finite populations, environmental change, mutation, and modifier dissociation all limit the

lifetime of nearly neutral competition and therefore round the deterministic singularity.

None of these effects changes the central geometric observation. Opposing component fitness differences can cancel in total while leaving a finite negative genomic association. As the total selective difference approaches zero, that association persists for an increasingly long time. Maximum configurational entropy, maximally favorable covariance, and critical slowing down therefore coincide.

The broader implication is that statements about whether sex “increases entropy” are incomplete unless the relevant state space and scale are specified. Sex can be most strongly favored at maximum entropy in one space precisely because substantial genomic order remains available to be destroyed in another. In populations containing many genotypes, generic near-critical events reduce to competition between two nearly degenerate leading backgrounds, preserving both the positive singularity and its characteristic square-root entropy scaling.

Declaration of competing interest

The author declares no competing interests.

Data and code availability

No empirical data are analyzed in this study. Julia code used to generate the numerical results and figures will be made publicly available with the published article.

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